

Probability Notes

Random Variables, Expectation, and Key Distributions

Dhruv Parsewar

1. Random Variables and Expectation

- **Discrete random variables** $\rightarrow$ probability mass function (pmf):

$$p(x) = P(X = x)$$

- **Continuous random variables** $\rightarrow$ probability density function (pdf):

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

Expectation ($E[X]$) $\rightarrow$ **probability weighted average**

$$\text{Discrete} \rightarrow E[X] = \sum x_i p_i$$

$$\text{Continuous} \rightarrow E[X] = \int x f(x) dx$$

2 properties of expectation:

1. **Linearity:** $E[aX + bY] = aE[X] + bE[Y]$
2. **Indicator trick:** for event A , $E[1_A] = P(A)$
(*random variable when its value is 1 if it happens, otherwise 0*)

2. Variance and Key Properties

Variance: $Var(X) = E[X^2] - (E[X])^2$

Derivation:

$$\begin{aligned} Var(X) &= E[(X - \mu)^2] \quad (\mu \rightarrow \text{mean}) \\ &= E[(X - E[X])^2] \\ &= E[X^2 - 2XE[X] + (E[X])^2] \quad (\text{Note: } E[X] \text{ is constant}) \\ &= E[X^2] - 2E[XE[X]] + (E[X])^2 \\ &= E[X^2] - 2E[X]E[X] + (E[X])^2 \\ &= E[X^2] - (E[X])^2 \end{aligned}$$

Standard Deviation: $\sqrt{Var(X)}$ (By defn)

Properties of Variance

1. **Constant:** $Var(aX + b) = a^2 Var(X)$

$$\begin{aligned} Var(aX + b) &= E[((aX + b) - E[aX + b])^2] \quad (\text{By defn}) \\ &= E[((aX + b) - (aE[X] + b))^2] \\ &= E[(aX - aE[X])^2] \\ &= E[a^2(X - E[X])^2] \\ &= a^2 Var(X) \end{aligned}$$

2. **Sum:** $Var(X + Y) = Var(X) + Var(Y) + 2Cov(X, Y)$

$$\begin{aligned}
 Var(X + Y) &= E[((X + Y) - (E[X] + E[Y]))^2] \\
 &= E[((X - E[X]) + (Y - E[Y]))^2] \\
 &= Var(X) + Var(Y) + 2E[XY + E[X]E[Y] - XE[Y] - YE[X]] \\
 &= Var(X) + Var(Y) + 2\underbrace{(E[XY] - E[X]E[Y])}_{Cov(X,Y) \text{ (By defn)}} \\
 &= Var(X) + Var(Y) + 2Cov(X, Y)
 \end{aligned}$$

3. **If X and Y are independent:**

$$Var(X + Y) = Var(X) + Var(Y) \quad \text{i.e. } Cov(X, Y) = 0$$

coz for independent X and Y , $E[XY] = E[X]E[Y]$

3. Some Common Probability Distributions

Discrete Distributions

1) Bernoulli

pmf $\rightarrow P(X = 1) = p$ (success), $P(X = 0) = 1 - p$

• **mean** $\rightarrow E[X] = p$

$$\begin{aligned}
 E[X] &= \sum x \cdot p(x) \quad (\text{coz discrete}) \\
 &= 1 \cdot p + 0 \cdot (1 - p) \\
 &= p
 \end{aligned}$$

• **variance** $\rightarrow Var(X) = p(1 - p)$

$$\begin{aligned}
 Var(X) &= E[X^2] - (E[X])^2 \\
 &= (1^2 \cdot p + 0^2 \cdot (1 - p)) - p^2 \\
 &= p - p^2 \\
 &= p(1 - p)
 \end{aligned}$$

2) Binomial

pmf $\rightarrow P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$ ($n \rightarrow$ total trials, $k \rightarrow$ success)

• **mean** $\rightarrow E[X] = np$

$$\begin{aligned}
 E[X] &= \sum_{k=0}^n k \cdot \binom{n}{k} p^k (1 - p)^{n-k} \\
 &= \sum_{k=1}^n n \binom{n-1}{k-1} p^k (1 - p)^{n-k} \\
 &= np \sum_{k=1}^n \binom{n-1}{k-1} p^{k-1} (1 - p)^{n-k} \\
 &= np(p + (1 - p))^{n-1} \\
 &= np
 \end{aligned}$$

- **variance** $\rightarrow Var(X) = np(1-p)$

Using $E[X^2] - (E[X])^2$, we calculate $E[X(X-1)] + E[X]$:

$$\begin{aligned} E[X(X-1)] &= \sum_{k=0}^n k(k-1) \binom{n}{k} p^k (1-p)^{n-k} \\ &= \sum_{k=2}^n n(n-1) \binom{n-2}{k-2} p^k (1-p)^{n-k} \\ &= n(n-1)p^2 \sum_{k=2}^n \binom{n-2}{k-2} p^{k-2} (1-p)^{n-k} \\ &= n(n-1)p^2 (p+1-p)^{n-2} \\ &= n(n-1)p^2 \end{aligned}$$

So, $Var(X) = n(n-1)p^2 + np - (np)^2 = np(1-p)$

OR Use indicator trick!!!

$X = I_1 + I_2 + \dots + I_n$ where I_i represents event for i^{th} trial.

$$E[X] = \sum_{i=1}^n E[I_i] = np \quad (\text{coz } E[I_i] = p \text{ for each } 1 \leq i \leq n)$$

$$Var(X) = \sum_{i=1}^n Var(I_i) = np(1-p) \quad (\text{coz } Var(I_i) = p(1-p) \text{ for each } 1 \leq i \leq n)$$

3) Poisson

Binomial to Poisson $\rightarrow$ imagine taking n trials and multiplying them toward infinity, while prob. p of each trial shrinks towards zero, such that avg rate of success ($n \times p$) remains const. $\rightarrow$ this constant is λ (rate).

pmf $\rightarrow P(X=k) = \frac{e^{-\lambda} \lambda^k}{k!}$ for $k \geq 0$

Binomial dist. $\rightarrow$ Binomial thm. Poisson dist. $\rightarrow$ Taylor series (expansion of e^x): $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$

- **mean** $\rightarrow E[X] = \lambda$

$$E[X] = \sum_{k=0}^{\infty} k \cdot \frac{e^{-\lambda} \lambda^k}{k!} = \sum_{k=1}^{\infty} \frac{e^{-\lambda} \lambda^k}{(k-1)!} = \lambda e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} = \lambda e^{-\lambda} e^{\lambda} = \lambda$$

- **variance** $\rightarrow Var(X) = \lambda$

$$Var(X) = E[X^2] - (E[X])^2 \quad (E[X^2] \text{ computed using } E[X(X-1)] + E[X])$$

$$E[X(X-1)] = \sum_{k=0}^{\infty} k(k-1) \frac{e^{-\lambda} \lambda^k}{k!} = \lambda^2 e^{-\lambda} \sum_{k=2}^{\infty} \frac{\lambda^{k-2}}{(k-2)!} = \lambda^2 e^{-\lambda} e^{\lambda} = \lambda^2$$

$$Var(X) = (\lambda^2 + \lambda) - \lambda^2 = \lambda$$

For Poisson's dist. both mean and variance are λ .

Continuous Distributions

Total area = 1 ofc

1) Uniform

pdf $\rightarrow f(x) = \frac{1}{b-a}$ for $a \leq x \leq b$ ($a \rightarrow$ lower limit, $b \rightarrow$ upper limit)

- **mean** $\rightarrow E[X] = \frac{a+b}{2}$

$$\int_a^b x \cdot \frac{1}{b-a} dx = \frac{1}{2(b-a)} [x^2]_a^b = \frac{1}{2(b-a)} (b^2 - a^2) = \frac{a+b}{2}$$

- **variance** $\rightarrow Var(X) = \frac{(b-a)^2}{12}$

$$Var(X) = E[X^2] - (E[X])^2$$

$$E[X^2] = \int_a^b x^2 \cdot \frac{1}{b-a} dx = \frac{1}{3(b-a)} [x^3]_a^b = \frac{b^3 - a^3}{3(b-a)} = \frac{b^2 + a^2 + ab}{3}$$

$$\begin{aligned} Var(X) &= \frac{b^2 + a^2 + ab}{3} - \left(\frac{a+b}{2} \right)^2 \\ &= \frac{b^2 + a^2 + ab}{3} - \frac{a^2 + b^2 + 2ab}{4} \\ &= \frac{4a^2 + 4b^2 + 4ab - 3a^2 - 3b^2 - 6ab}{12} \\ &= \frac{a^2 + b^2 - 2ab}{12} = \frac{(b-a)^2}{12} \end{aligned}$$

2) Normal (or Gaussian)

Here mean (μ) and variance (σ^2) are the parameters for defining the distribution. (Bell curve)

$$\text{pdf} \rightarrow f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- **mean** $\rightarrow E[X] = \mu$

$$\begin{aligned} E[X] &= \int_{-\infty}^{\infty} x \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx \\ \text{put } z &= \frac{x-\mu}{\sigma} \text{ i.e. } x = z\sigma + \mu, dx = \sigma dz \\ &= \int_{-\infty}^{\infty} \frac{z\sigma + \mu}{\sqrt{2\pi\sigma^2}} e^{-\frac{z^2}{2}} \sigma dz \\ &= \frac{\sigma}{\sqrt{2\pi}} \underbrace{\int_{-\infty}^{\infty} z e^{-\frac{z^2}{2}} dz}_{\text{odd fn} \rightarrow 0} + \mu \underbrace{\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz}_{\text{Standard Normal dist, AUC}=1} \\ &= 0 + \mu(1) = \mu \end{aligned}$$

- **variance** $\rightarrow Var(X) = \sigma^2$

$$Var(X) = E[(X - \mu)^2] = \int_{-\infty}^{\infty} (x - \mu)^2 \frac{1}{\sqrt{2\pi}\sigma^2} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

$$\text{put } z = \frac{x - \mu}{\sigma} \text{ i.e. } x = z\sigma + \mu, dx = \sigma dz$$

$$= \int_{-\infty}^{\infty} \frac{z^2 \sigma^2}{\sqrt{2\pi}\sigma^2} e^{-\frac{z^2}{2}} \sigma dz$$

$$= \sigma^2 \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} z^2 e^{-\frac{z^2}{2}} dz$$

Use by parts:

$$= \sigma^2 \frac{1}{\sqrt{2\pi}} \left(\left[-ze^{-z^2/2} \right]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} e^{-\frac{z^2}{2}} dz \right)$$

$$= \sigma^2 \left(0 + \underbrace{\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz}_{\text{Standard Normal Dist.}} \right)$$

$$= \sigma^2$$

4. Independence, Covariance, and Conditional Probability

Independence and Covariance

- A and B are independent if $P(A \cap B) = P(A)P(B)$.
- Random variables X, Y are independent if their joint distribution factors:

$$f(x, y) = f_X(x) \cdot f_Y(y)$$

(Marginal probability dist $X = x$ and ignore Y)

- For independent X & $Y \rightarrow E[XY] = E[X]E[Y]$ and $Var(X + Y) = Var(X) + Var(Y)$.

Covariance measures **dirⁿ** of dependence

$$Cov(X, Y) = E[XY] - E[X]E[Y]$$

The correlation coefficient normalises to $[-1, 1]$:

$$\rho(X, Y) = \frac{Cov(X, Y)}{\sigma_X \sigma_Y}$$

- $\rho = 0 \rightarrow$ uncorrelated but not necessarily independent.
- $\rho = 1 \rightarrow$ perfect +ve linear relationship (converse is true tho).
- $\rho = -1 \rightarrow$ perfect -ve linear relationship.

Conditional Probability and Bayes' Theorem

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Bayes' Thm:

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$